[redacted]

Literacy Learning as a Social Practice

Warner School for Education

University of Rochester

Spring 2025

Unit Context:

This unit was built and designed to be delivered to 4th grade students attending the Durand-Eastman Intermediate School. A public school located within the East Irondequoit School District, in Rochester, New York in a large suburb setting. The classroom has about 58% minority enrollment, with a student-teacher ratio of 11:1. The student population is made up of about 47% female and 53% male students. Approximately 42.4% of the students identify as White, 25.2% Hispanic/Latino, 20.0% Black, and about 12.6% multi-racial. According to the most recent assessments only about 28% of students scored at or above the proficiency level for math, and 20% scored at or above the proficiency level for reading. Most students are below expectations in both math and reading. On top of this the school enrolls about 67% economically disadvantaged students. This unit is designed for face-to-face multimodal and interdisciplinary instruction relying on hands on activities with digital resources to back up student's consistent research throughout. This unit assumes each student has access to a district-issued computer and internet within the classroom.

This unit integrates literacy and critical thinking skills focusing on science, particularly renewable energy and environmental advocacy. This unit offers support with complex concepts using visual aids, group discussions, and hands-on learning. This is important as many of the students in this setting are below proficiency in both reading and math. By providing multimodal opportunities the hope is that student comprehension and engagement will be enhanced, and language further developed. This unit further taps into students' lived experiences and cultivating relevance and empowerment by relating science content to advocacy and real-world issues. This is crucial in a diverse, economically disadvantaged school environment as it allows all students to feel seen and heard furthering their literacy skills.

Unit Description and Purpose:

Before beginning this mini-unit, students in my classroom have already explored science concepts about energy in a previous full unit. Within this, they understand that energy is present in moving objects and that faster motion means more energy. They also have adequate understanding of the fact that energy can be transferred in different forms and that when objects collide energy can shift and/or transform. Students have also begun to develop an awareness of how energy is used in everyday life through the conversion of stored energy into motion, light, heat, or sound. This prior knowledge will set students up for success as they dive into the mini unit *Renewable Energy and Environmental Advocacy,* answering the essential question:

"How do the choices we make about energy help/hurt the Earth and the people in it?"

This unit is designed to be interdisciplinary, working towards addressing both science and literacy standards. Students will investigate the different sources of energy we use, the differences between renewable and nonrenewable resources, and how energy choices impact people and the environment. This inquiry is important as it brings the common statement "actions have consequences" into a broader context. Instead of connecting everyday choices to immediate consequences and/or favorable results it offers a broader perspective that everyday choices can have global consequences, making the status of the environment personal for students. For 4th graders learning how their actions can benefit other people, and their community help them see themselves as change seekers. Students learning that they can make a difference and help others provide a connection to literacy beyond just skill based, which is the key to education.

This unit attempts to acknowledge student diversity and the difficulties that some cultures and communities face by situating it in real-world settings. By combining research, debate, and advocacy writing, it encourages literacy engagement while advancing communication and content comprehension. In the end, this unit advances equity by empowering all students, particularly those from underrepresented backgrounds, to see themselves reflected in the curriculum and then to act and speak up to help create a more sustainable future.

Literacies Framework:

The unit will address each layer of the Culturally and Historically Responsive Literacy Framework in the following ways:

Joy: Joy is woven throughout this unit in a variety of ways that are both creative and empowering. Students will get to experience the excitement of street art, design their own artistic posters, examine 3D structures, and take inspiration from other youth changemakers. They will also get to build and test their own windmills.

Identity: This unit empowers students to explore their identities as environmental advocates, engineers, change seekers, and changemakers. Through examples like Chispa activists, Little Miss Flint, William Kamkwamba, and the creation of their own personal ideas with connections to their current lives, students will see how young people from a variety of diverse backgrounds make a real impact. By naming their personal "spark" and designing solutions, students will reflect on their own values, passions, and community roles.

Skills: Students will develop a range of literacy and STEM skills throughout this unit including:

1. Reading and analyzing informational texts.
2. Engaging in a range of collaborative discussions to build understanding.
3. Conducting research and recording findings in engineering notebooks.
4. Writing persuasive letters with clear claims, evidence, and calls to action.
5. Applying the engineering design process to create windmills

All the previously states skills align with the NYS Science and ELA standards, supporting reading, writing, speaking, and inquiry.

Intellect: This unit works to deepen student understanding of renewable energy, environmental justice, and engineering design. Through the examination of real-world problems and activist solutions students will work to connect scientific concepts (wind power/natural resources) to social and environmental challenges, building meaningful knowledge that links science, advocacy, and innovation.

Criticality: Students will engage in critical thinking about power, equity, and justice through the Flint water crisis and analyzing William Kamkwamba's story. Through both of these real stories' students will look at how access to clean energy and resources often affects marginalized communities. They will analyze who has power in environmental decisions and consider how youth voices can push for change (further recognizing that renewable energy is both a scientific and justice-based issue)

Lesson Plans:

Name of Lesson: Be an Inventor Like Edison!	**Grade Level: 4th Grade**
Teacher: Cora Rose	
Length of Lesson: 60 minutes	
Students' Identities and Background: The classroom has about 58% minority enrollment, with a student-teacher ratio of 11:1. The student population is made up of about 47% female and 53% male students. Approximately 42.4% of the students identify as White, 25.2% Hispanic/Latino, 20.0% Black, and about 12.6% multi-racial. According to the most recent assessments only about 28% of students scored at or above the proficiency level for math, and 20% scored at or above the proficiency level for reading. Most students are below expectations in both math and reading. On top of this the school enrolls about 67% economically disadvantaged students.	
Learning Goals	**Joy:** Students will watch videos of art that utilize energy making notes of differences, feelings, and/or thoughts about each piece. **Identities:** Students will connect their own use of energy to broader environmental impacts. **Skills:** Students will engage in the following NYS Science and Reading Standards. Science: o 4-ESS3-1: Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment. E.L.A: o 4SL1: Engage effectively in a range of collaborative discussions with diverse partners, expressing ideas clearly, and building on those of others o 4W6: Conduct research to answer questions, including self-generated questions, and to build knowledge through investigating multiple aspects of a topic Objectives: 1. Students will be able to define and provide examples of non-renewable energy **Intellect:** Students will begin to evaluate how different energy systems can impact people and the planet, developing ideas for how we might do things differently. **Criticality:** Students will begin to question how different energy choices affect the environment and equity.

Layered Texts	Renewable Energy Art Video → https://youtu.be/agYO1oXuiEY Non-Renewable Energy Art Video → https://www.youtube.com/watch?v=nX6C6xY86-c Thomas Edison's Notebook Excerpts (Appendix A) EIA Kids Energy Sources → https://www.eia.gov/kids/energy-sources/gasoline/
Vocabulary & Concepts	Vocabulary: o Renewable o Non-Renewable o Fossil Fuels o Energy Source o Natural Resources Concepts: o Sustainability
Student Spark	To get students excited about renewable energy, students will begin the lesson by watching two sort video clips showcasing real-world moving art installations. The first piece titled "LUCEA" by Anthony Howe is powered by wind. The second piece titled "Can't Help Myself" by Sun Yuan and Peng Yu is powered by electricity via a high voltage cable utilizing the cities power grid from natural gas and/or fuel oil. In other words, one artwork is powered by a renewable energy source, and one is powered by a non-renewable one. As students view the two pieces, they are encouraged to jot down any thought, feeling, or difference between the two on a blank sheet of paper. After students view the two pieces the whole class will engage in a brief discussion comparing the two. - What energy source is powering each piece? - Which one do you think is more reliable, and why?
Body of Lesson	**Intro:** (12 minutes) 1. The teacher turns lights on and off slowly a couple of times in a row. Quickly asking the class what the difference between the two are. Following the statement of light and dark the teacher will lead a discussion on the concept of energy, providing a quick overview of: - What is energy? - Where do you see it/use it in your day-to-day life? (5 minutes) 2. Teacher will segway into watching the two videos and having a discussion described in the "Student Spark Section." (7 minutes) **Body:** (48 minutes) 3. The teacher will introduce the concept of an Engineering Notebook that each student will keep all semester. To introduce this concept the teacher will show at the start of each class excerpts of different engineers' notebooks. To start excerpts from Thomas Edison's idea book will be shown (Appendix A). - The teacher will provide background that Thomas Edison invented the light bulb and will ask students to point out what he included in his notebook (pictures, words, etc.) - Teacher will pass out notebooks stating that students will use this for all their notes, reflections, and designs throughout the unit. - Students will glue their energy art video thoughts to their first page. - The terms renewable and non-renewable energy will be introduced using a simple anchor chart at the front of the room:

	 - In pairs students will use the EIA Kids Energy Sources website to take notes on specifically non-renewable energy sources. In their notebooks they will record: o What is it? Examples? o How is it used? o Pros/Cons of using nonrenewable sources? (Note: Each student must fill out their own notebook, but collaboration is highly encouraged) - Whole group discussion will follow, adding the answers to the non-renewable section of the anchor chart.
Closure	Students will be asked to complete an Exit Ticket in their notebook: - One thing I learned about energy today is..... - One question I still have is...
Assessment	1. Students will be able to define and provide examples of non-renewable energy o Informal observation during whole group sharing o Engineering notebook notes on non-renewable energy

Name of Lesson: Renewable Energy!	**Grade Level: 4th Grade**
Teacher: Cora Rose	
Length of Lesson: 2 60-minute periods (First class students will go through the introduction activity and 2 stations. The second-class students will do through the remaining 3 stations they haven't gotten to yet).	
Students' Identities and Background: The classroom has about 58% minority enrollment, with a student-teacher ratio of 11:1. The student population is made up of about 47% female and 53% male students. Approximately 42.4% of the students identify as White, 25.2% Hispanic/Latino, 20.0% Black, and about 12.6% multi-racial. According to the most recent assessments only about 28% of students scored at or above the proficiency level for math, and 20% scored at or above the proficiency level for reading. Most students are below expectations in both math and reading. On top of this the school enrolls about 67% economically disadvantaged students.	

Learning Goals	**Joy:** Students will experience the joy of discovery and invention through working in hands-on stations. The integration of engineering notebooks, station rotations, and interactive group work allows them to experience the excitement of collaboration. **Identities:** Students will explore how inventors and engineers from diverse backgrounds (in this case Alfred L. Cralle) have contributed to solving real world problems. On top of this students will connect the energy forms that they learn about in class to the energy sources that they currently use at home. **Skills:** Students will engage in the following NYS Science and Reading Standards. Science: ○ 4-ESS3-1: Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment. E.L.A: ○ 4SL1: Engage effectively in a range of collaborative discussions with diverse partners, expressing ideas clearly, and building on those of others ○ 4W6: Conduct research to answer questions, including self-generated questions, and to build knowledge through investigating multiple aspects of a topic Objectives: 1. Students will be able to identify and explain five types of renewable energy 2. Students will be able to state the pros and cons of renewable and non-renewable energy sources **Intellect:** Students will deepen their understanding of energy systems and sustainability by investigating how different renewable energy sources work, along with their benefits and challenges and their global impacts. **Criticality:** Students will begin to question how different energy choices affect the environment and equity.
Layered Texts	Alfred L. Cralle Engineering Excerpts (Appendix B) Solar Station: 1. https://youtu.be/odiHDruGr_o 2. https://kids.britannica.com/kids/article/solar-energy/433607#:~:text=Solar%20cells%20use%20the%20Sun's,provide%20electricity%20for%20whole%20buildings. Wind Station: 1. https://www.dailymotion.com/video/x6qa1cl 2. https://www.energy.gov/eere/wind/explore-wind-turbine Geothermal Station: 1. https://youtu.be/mCRDf7QxjDk 2. https://www.energy.gov/eere/geothermal/geothermal-basics Hydropower Station: 1. https://www.nationalgeographic.com/environment/article/hydropower#:~:text=How%20hydropower%20works,a%20waterway%20without%20a%20dam. Biomass Station: 1. https://youtu.be/7cCSV0IO4zE 2. https://www.eia.gov/kids/energy-sources/biomass/
Vocabula Concept	Vocabulary: ○ Renewable ○ Non-Renewable ○ Energy Source ○ Natural Resources ○ Fossil Fuels Concepts: Sustainability
Student Spark	To continually re-iterate throughout the semester the purpose of the engineering notebook each class will start with looking at/examining different inventors engineering notebooks/design sketches. The first one showed was Thomas Edison. For this lesson the teacher will project images of Alfred L. Cralle's patent sketch for the mechanical ice cream scoop (Appendix B). Students will briefly be told that at the end of this unit they will be working to design and create their own renewable energy source (which is what they will be learning about today). These will then be posed the following questions to be answered in an open class discussion:

1. What do you notice about how Alred L. Cralle recorded his ideas?
2. Why do you think inventors keep notebooks?
3. What everyday problems would you solve with a new invention?

This exploration aims to allow students to see themselves as inventors.

Body of Lesson

Intro: Student Spark Section (7 minutes)

Body: Students will be split up into 5 equal groups. (53 minutes first day and about 50 minutes second day)

- Each group will rotate through 5 stations (solar energy, wind energy, hydropower, geothermal energy, and biomass).
- At each station each group will work through the following texts/videos provided at each spot in the classroom (links in the layered text section)
- In their engineering notebooks students will create a slightly altered version of a KWL chart with the following columns filling out each part of the table within their groups

	What is this Energy Source?	How does it work?	Benefits?	Challenges?
Wind Energy				
Hydropower				
Geothermal Energy				
Biomass				
Solar Energy				

- After the lessons the group will convene as a whole on both days of this lesson, answering the following questions:
  1. Which renewable energy surprised you the most?
  2. Which one would work best in your neighborhood?

Closure

For the last 10 minutes of the second-class students will participate in a "Renewable Energy Relay." In teams they will take turns coming up to a large chart at the front of the classroom labelled with the 5 types of renewable energy. Each team will have to add one NEW fact observation or question they have about each station. After the relay, the class will quickly review the whole chart together.

Assessment

1. Students will be able to identify and explain five types of renewable energy
- The completion of modified student KWL charts at each station and group
2. Students will be able to state the pros and cons of renewable and non-renewable energy sources
- This is evident in the homework they will receive at the end of the second day of this lesson. Again, in their engineering notebooks they must list and answer the following:
  1. List and describe the two items they found (or just the non-renewable one if they couldn't find a renewable one).
  2. State whether each item was powered by a renewable or non-renewable energy source
  3. Write two sentences for each item. One explaining a pro of the energy source and one explaining a con. The teacher will look at all notebooks at the beginning of the following day to ensure learning occurred.

Name of Lesson: Wind Energy Case Study	**Grade Level: 4th Grade**
Teacher: Cora Rose	
Length of Lesson: 60 Minutes	

Students' Identities and Background: The classroom has about 58% minority enrollment, with a student-teacher ratio of 11:1. The student population is made up of about 47% female and 53% male students. Approximately 42.4% of the students identify as White, 25.2% Hispanic/Latino, 20.0% Black, and about 12.6% multi-racial. According to the most recent assessments only about 28% of students scored at or above the proficiency level for math, and 20% scored at or above the proficiency level for reading. Most students are below expectations in both math and reading. On top of this the school enrolls about 67% economically disadvantaged students.	
Learning Goals	**Joy:** Students will be able to engage in collaboration and creativity through the creation of a mini poster. **Identities:** Students will explore how William Kamkwamba's cultural background and lived experiences shaped his ability to solve a community problem. They will then reflect on how their own identities, ideas and environments can influence creative problem-solving and community action. **Skills:** Students will engage in the following NYS Science and Reading Standards. Science: o 4-ESS3-1: Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment. E.L.A: o 4SL1: Engage effectively in a range of collaborative discussions with diverse partners, expressing ideas clearly, and building on those of others o 4W6: Conduct research to answer questions, including self-generated questions, and to build knowledge through investigating multiple aspects of a topic o 4R1: Locate and refer to relevant details and evidence when explaining what a text says explicitly/implicitly and make logical inferences Objectives: 1. Students will identify wind energy as a type of renewable energy and explain why it is important for sustainable living 2. Students will use textual evidence and illustrations to make predictions, ask questions, and infer meaning during and after reading *The Boy Who Harnessed the Wind* 3. Students will discuss the inequities in global access to electricity **Intellect:** Students will continue to build content knowledge around renewable energy (specifically wind) and how it can serve as a sustainable solution to real-world challenges. **Criticality:** Students will begin to truly examine the issues of equity and access to energy, using William's story to spark reflection on power and fairness of energy. Through some guided discussion students will consider why some communities have access to reliable electricity and why some do not. Further beginning to understand how renewable energy can be a tool to solve this issue.
Layered Texts	Guillermo González Camarena Notebook Excerpt (Appendix C) https://flowcharts.llnl.gov/commodities/energy --> Intro/student spark activity Short Clip of End of the Boy Who Harnessed the Wind Movie → **https://www.youtube.com/watch?v=lrucHzfCAB0&t=24s** Teacher Read-Aloud: The Boy Who Harnessed the Wind Young Readers Edition → https://www.youtube.com/watch?v=5GDPsFaYIxI
Vocabulary & Concepts	Vocabulary: o Windmill o Wind Turbine o Energy o Electricity Concepts: o Sustainability o Engineering Solutions o Equity in Access to Energy

Student Spark	Students in pairs will be asked to explore the following website for 3 minutes: https://flowcharts.llnl.gov/commodities/energy In a class discussion students will work towards answering the following questions: 1. Which energy sources are the biggest? Smallest? 2. Are they renewable? Non-Renewable? 3. Why do you think renewable energy sources make such a small contribution? The teacher will state that we are going to shift gears towards specifically wind energy now. 4. What is wind energy's energy trend? 5. Do you think wind energy will become a bigger part of this chart in the future? Should it?

Body of Lesson	**Intro:** Students will start by examining the Engineering Notebook of the day. Todays will be Guillermo González Camarena invention of the color television system (Appendix B). After looking at his notebook excerpt students will discuss the following questions:
	1. Any similarities between Camarenas and Edisons/Cralle's? Differences?
	2. What about your current engineering notebook? What should you keep doing? What should you add/change? (5 minutes)
	Student Spark Section (7 minutes)
	Watch short clip of movie asking students to just observe what they think is occurring (3 minutes)
	Body: (48 minutes)
	Before Reading the Story:
	- The teacher will show students the cover of The Boy Who Harnessed the Wind and ask:
	  ○ Why do you think there is a windmill in the background?
	  ○ What is the purpose of a windmill?
	- During Reading:
	  ○ (p. 1) I wonder if William's life is different than ours?
	  ○ (p. 6) How did the illustrator let us know that there was a drought
	  ○ (p. 8-9) What hardships did William suffer as a result of the flood?
	  ○ (p. 10-11) How did going to the library help William? What was his dream for his people? Why do you think he used a windmill instead of something else? (His village was a windy area)
	  ○ (p. 12) Why do you think the authors said the windmill was a weapon? How can renewable energy be powerful for people?
	  ○ (p.14) I think it's interesting that the authors used African words in this story.
	  ○ (p. 25) I wonder what the authors meant when they say, "William knew he had just begun."
	- After reading:
	  ○ What problems did William solve with his windmill?
	  ○ Do you think it's fair that some communities have lots of electricity and others have little to none? Why do you think this happens?
	  *(Students will jot down thoughts in their engineering notebooks) *
	Students will split into groups of 3-4 to create a mini poster. Its requirements include:
	- State William's problem
	- State William's solution
	- How renewable energy (wind) helped
	- One idea for how we could use renewable energy in our own community

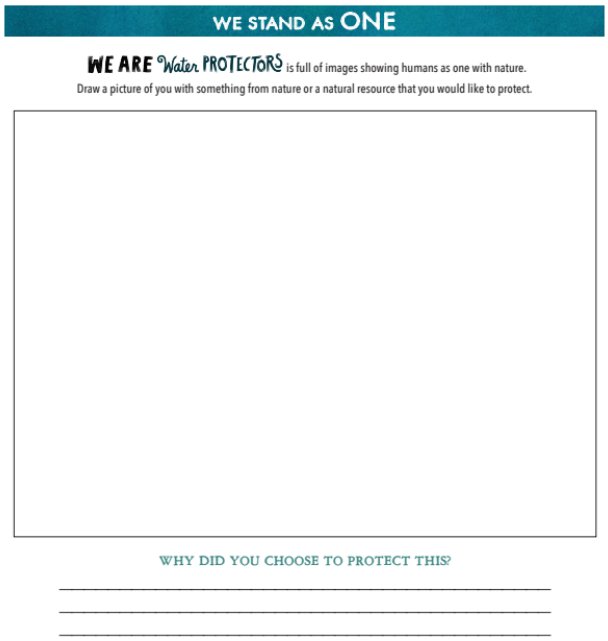
Closure	Following the creation of the students' mini posters students will perform a gallery walk of each other's posters. As they walk around, they will write down comments, questions, or drawings that come to mind while viewing their fellow students' posters.

Assessment	1. Students will identify wind energy as a type of renewable energy and explain why it is important for sustainable living 2. Students will use textual evidence and illustrations to make predictions, ask questions, and infer meaning during and after reading *The Boy Who Harnessed the Wind* - Students will create a mini poster addressing the questions stated prior. Will be graded using the simple rubric of: 1. Does it answer all required questions? 2. Does it use details from the story and/or illustrations? 3. Creativity and effort 3. Students will discuss the inequities in global access to electricity - In their engineering notebooks students will be required to write at least 3 sentences regarding their thoughts on this topic and what they gathered from the class discussion.

Name of Lesson: Environmental Advocacy: Voices for Renewable Energy	**Grade Level: 4th Grade**
Teacher: Cora Rose	
Length of Lesson: 2 60 Minute lessons (The first lesson will include the Street Art Activity, We Are Water Protectors, and the Chispa Activity. The second lesson will include the Little Miss Flint activities and the advocacy writing)	
Students' Identities and Background: The classroom has about 58% minority enrollment, with a student-teacher ratio of 11:1. The student population is made up of about 47% female and 53% male students. Approximately 42.4% of the students identify as White, 25.2% Hispanic/Latino, 20.0% Black, and about 12.6% multi-racial. According to the most recent assessments only about 28% of students scored at or above the proficiency level for math, and 20% scored at or above the proficiency level for reading. Most students are below expectations in both math and reading. On top of this the school enrolls about 67% economically disadvantaged students.	
Learning Goals	**Joy:** Students will experience joy by expressing their passion for the environment through creative art and writing, finding empowerment in discovering their "spark" and realizing that their voices have the power to create real change. **Identities:** Students will explore and reflect on their personal connections to environmental issues by identifying their own "spark" for protecting the Earth, drawing from both their lived experiences and cultural/community values. In addition to this, students will also recognize and appreciate the contributions of youth activists from diverse racial and cultural backgrounds (Indigenous, Latino, and Black communities) **Skills:** Students will engage in the following NYS Science and Reading Standards. Science: o 4-ESS3-1: Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment. E.L.A:

	○ 4SL1: Engage effectively in a range of collaborative discussions with diverse partners, expressing ideas clearly, and building on those of others ○ 4W6: Conduct research to answer questions, including self-generated questions, and to build knowledge through investigating multiple aspects of a topic ○ 4R1: Locate and refer to relevant details and evidence when explaining what a text says explicitly/implicitly and make logical inferences Objectives: 1. Students will research different environmental challenges through real-world activist examples. 2. Students will obtain and combine information from multiple texts to explain how environmental actions can affect natural resources and the Earth. **Intellect:** Students will deepen their understanding of how renewable and non-renewable energy sources connect/impact environmental issues worldwide. Students will also gain knowledge about how youth advocacy and collective actions can protect and restore natural resources. **Criticality:** Students will examine issues of environmental injustice, such as the Flint Water Crisis and pollution in marginalized communities, discussing why certain groups are disproportionately impacted and how advocacy can challenge inequity.
Layered Texts	Street Art 1. By Dr. Love at Upfest 2015 in Bristol, England 2. Conditional Love by SaRx We are Water Protectors by Carol Lindstrom → https://www.youtube.com/watch?v=jqWTouyaQ-8 Chispa Websites: 1. https://chispaaz.org/ 2. https://chispalcv.org/state/maryland/ Little Miss Flint: Letter to President:

	Mr. President, *Hello my name is Mari Copeny and I'm 8 years old, I live in Flint, Michigan and I'm more commonly known around town as "Little Miss Flint". I am one of the children that is effected by this water, and I've been doing my best to march in protest and to speak out for all the kids that live here in Flint. This Thursday I will be riding a bus to Washington, D.C. to watch the congressional hearings of our Governor Rick Snyder. I know this is probably an odd request but I would love for a chance to meet you or your wife. My mom said chances are you will be too busy with more important things, but there is a lot of people coming on these buses and even just a meeting from you or your wife would really lift people's spirits. Thank you for all that do for our country. I look forward to being able to come to Washington and to be able to see Gov. Snyder in person and to be able to be in the city where you live.* *Thank You* *Mari Copeny* Story of Little Miss Flint: https://www.rebelgirls.com/podcast/mari-copeny-water-for-flint
Vocabulary & Concepts	Vocabulary: o Advocacy o Renewable Energy o Environmental Justice o Pollution Concepts: Youth Activism Community Empowerment
Student Spark	After showing the two examples of street art shown in the layered text section the teacher will lead the following class discussion: o What do you notice first? o What message do you think the artist is trying to send? o How do the pictures make you feel? o Why might someone choose art (instead of just words) to walk about the environment?
Body of Lesson	**Intro:** Student Spark Section (7 minutes) Teachers will state that we have learned what renewable and non-renewable energy sources are and that renewable energy helps the environment. Today we are going to look at how young people can use their voices to stand up for the environment **Body:** 1. Read-Aloud: We are Water Protectors Before: - This story is about standing up for clean water. Why might water need protection? After: - What problem did the water protectors face? - How did they work together? - With what we have talked about in regards to renewable and non-renewable energy why is using renewable energy better for the environment?

	Students will fill out the following worksheet and glue it in their engineering notebook.  WE STAND AS ONE WE ARE Water PROTECTORS is full of images showing humans as one with nature. Draw a picture of you with something from nature or a natural resource that you would like to protect. WHY DID YOU CHOOSE TO PROTECT THIS? _____ _____ _____ 2. Students will then research and explore Chispa (Latino activist) websites. In their engineering notebooks they will answer the following questions: o What is Chispa fighting for? o Why is it important? o What is your spark for helping the Earth? o How do you want to take action? Teacher will mention that Chispa means "spark" → will ask students in their engineering notebooks what their "spark" is for helping the environment along with a visual that they feel accurately portrays how they feel about the environment. _____ 3. Students will listen to the story of Little Miss Flint, participating in the following class discussion: o What happened in Flint? o How did Mari use her voice to make change? o What can you do if you see something unfair is happening? Students will examine Little Miss Flints letter to the president and will be provided with the instructions that they will begin to write a persuasive letter to the President about why renewable energy is important
Closure	To close the lesson, the teacher will facilitate a reflective circle where students will share their "spark" and one thing they learned from the activists they studied. The teacher will guide the students to connect their learning to the power of using their voices and actions to make a difference.
Assessment	1. Students will research different environmental challenges through real-world activist examples. - Will be assessed through what students write in their notebook entries. 2. Students will obtain and combine information from multiple texts to explain how environmental actions can affect natural resources and the Earth. - Students will be required to write a persuasive letter to a local leader about why renewable energy is important. The letter must include the following: - Who you are and what is your "spark"?

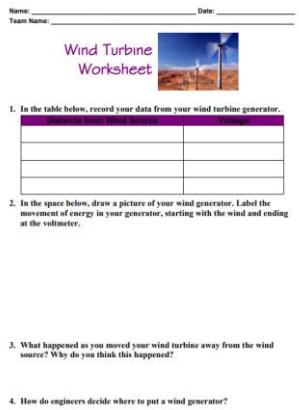
	- What problem you care about - Connection of letter from some parts of the We are Water Protectors, Chisa Activists, or Little Miss Flint stories/websites. - Why renewable energy matters (pros/cons to renewable/nonrenewable energy) - A specific idea/call to action

Name of Lesson: Engineering Challenge: Building a Windmill	**Grade Level: 4th Grade**
Teacher: Cora Rose	
Length of Lesson: 60 Minutes (if there is run over I will allot an additional 30 minutes the following class period for community implementation).	
Students' Identities and Background: The classroom has about 58% minority enrollment, with a student-teacher ratio of 11:1. The student population is made up of about 47% female and 53% male students. Approximately 42.4% of the students identify as White, 25.2% Hispanic/Latino, 20.0% Black, and about 12.6% multi-racial. According to the most recent assessments only about 28% of students scored at or above the proficiency level for math, and 20% scored at or above the proficiency level for reading. Most students are below expectations in both math and reading. On top of this the school enrolls about 67% economically disadvantaged students.	
Learning Goals	**Joy:** Students will experience joy through hands-on creation, teamwork, and seeing their windmills in action. **Identities:** Students will see themselves as capable inventors and engineers connecting their lived experiences to solutions for clean energy. **Skills:** Students will engage in the following NYS Science and Reading Standards. Science: o 4-ESS3-1: Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment. o 3-5-ETS1-1: Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost o E.L.A: o 4SL1: Engage effectively in a range of collaborative discussions with diverse partners, expressing ideas clearly, and building on those of others o 4W6: Conduct research to answer questions, including self-generated questions, and to build knowledge through investigating multiple aspects of a topic Objectives: 1. Students will use precise drawings to plan and communicate their original design ideas. 2. Students will be able to follow the steps of an engineering design process to create a functional windmill model. 3. Students will record observations **Intellect:** Students will deepen their understanding of renewable energy by exploring how wind turbines convert wind into usable energy. **Criticality:** Students will reflect on how renewable energy can reduce environmental harm and how creative solutions can challenge inequities.

Layered Texts	Sarah Goode Engineering Excerpt (Appendix D) Windmill instructions → https://www.teachengineering.org/activities/view/cub_energy2_lesson07_activity2
Vocabulary & Concepts	Vocabulary: ○ Wind Energy ○ Renewable Energy ○ Turbine ○ Blade Concepts: Engineering Design Process
Student Spark	The final engineering excerpt in this mini unit will feature Sarah Goode's invention and sketch of her cabinet bed (a piece of furniture that transforms a cabinet into a bed). Since this is the last example, students will be reminded that today is the day that all their previous observations from engineering notebooks will pay off (as they will get to create their own engineering sketch and design plan). Before starting the following discussion will be held: ○ What have you noticed as a common theme in all of the notebook excerpts (all will then be shown)? ○ What does this mean you should include in yours? ○ How much detail is used? ○ What is written down? What is excluded, if anything?
Body of Lesson	**Intro:** Student Spark (7 minutes) **Body:** Students will be provided with the following challenge (will be displayed on the board): **"Welcome to Windy Ridge, a small town high up on a grassy hilltop. The people here love their beautiful skies, big open fields, and the whooshing sound of the wind that never seems to stop.** **But there's a problem...** **Windy Ridge keeps running out of electricity. They've been using gas generators, but those are polluting the air and making people sick. The town wants to switch to clean, renewable energy that won't hurt the environment.** **There's no river nearby, and it hardly ever rains. Water power is out.** **But one thing Windy Ridge has *plenty* of... is wind!** **The townspeople are asking:** **Can we use the wind to power our lights, schools, and farms?** **They've called on you, a young inventor, to come up with a solution"** And materials (will be displayed on the board): • small DC toy motor; available online • 2 pieces of thin electrical wire with alligator clips, each about 50 cm or 20 inches long • rubber band • stiff ruler • cylindrical-shaped cork, at least 2 cm or ¾ inch in diameter; alternative to cork: Styrofoam ball • 4 paperclips

- scotch tape
- scissors
- 4 pieces of cardboard, each 3 x 5 cm

Using the following materials, the students will be instructed to write in their engineering notebooks the goal of the assignment, the materials required, and to create their own design for their windmill (drawing it as if they were inventors/creators (as they are)).
After the creation of their windmill's students will fill out the following worksheet:



Name: _____ Date: _____
Team Name: _____

Wind Turbine Worksheet

1. In the table below, record your data from your wind turbine generator.

Distance from Wind Source	Voltage

2. In the space below, draw a picture of your wind generator. Label the movement of energy in your generator, starting with the wind and ending at the voltmeter.

3. What happened as you moved your wind turbine away from the wind source? Why do you think this happened?

4. How do engineers decide where to put a wind generator?

(Note → This may go over into the next 30-minute time slot)
Students will then have the following discussion:

- What challenges did you face?
- How does this activity connect to real life?
- Why is it exciting and important to create clean energy solutions?

Closure

To close out the mini unit as a whole, student will join a Joy Circle where each student will take turns sharing one thing they felt proud of and one way they think their windmill could help people if it were real. To celebrate students as both inventors and environmental hero's students will take a class photo with their windmills to display in the classroom under the statement "Voices for Change."

Assessment

1. Students will use precise drawings to plan and communicate their original design ideas.
2. Students will be able to follow the steps of an engineering design process to create a functional windmill model.
3. Students will record observations

- For all of the objectives the end result of a working windmill will be looked at by the teacher as well as the design drawing. A full credit notebook entry will have complete sketch of the windmill with labelled parts, a material list, and the goal of the assignment.
- Lastly, to summarize the unit as a whole students will be required to answer the following question in their notebook:
  "Why is it exciting and important to create clean energy solutions?"
  - Their answers will be looked at to see if students met the following standard of: 4-ESS3-1: Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment.

Lesson Sample (From Lesson 4: Environmental Advocacy: Voices for Renewable Energy Persuasive Letter to Local Leader):

Note: All information obtained to create this advocacy letter came from what would be written in personal engineering notebooks. Everything from what the book inspired me to learn, details about the effects of both renewable and nonrenewable energy, and calls to action were talked about throughout the whole unit and students are required to put all pieces together in this advocacy letter.

Dear Mayor,

My name is Ms. Rose, and I am a 4th grade teacher at Durand Eastman Intermediate School. My "spark" for helping the Earth is my love of clean water. I love to swim, kayak, and paddleboard, and if our water were to get polluted, I would be devastated. Water also provides a home and drinking water for all plants and animals around the globe, and pollution could destroy them. Recently, my students and I read *We Are Water Protectors* by Carole Lindstrom. What I got out of this book is that we must all stand up and take action to protect Earth's water. The book showed me how powerful communities can be when we work together. It also reminded me that water is life, and without clean water, everything else is at risk.

Renewable Energy is one way that I believe we can protect our environment. By using clean sources of energy like solar and wind, we can reduce pollution and help keep our water and air safe. Specifically solar energy works to convert suns light energy into electricity. Its supply is endless in comparison to the limited fossil fuels which creates dangerous pollution that harms people, animals, and plants.

I am asking you to help bring more renewable energy to our community. One thing that I have been actively doing is making sure I turn my lights off when they are not in use. However, I think it would be a great idea to bring more renewable energy to the community. One idea is to install solar panels on our school buildings to save energy and teach students more about clean power. Another idea is to create a community program that helps families make their homes more energy efficient.

Thank you for your time and for working hard to keep our city safe and healthy!

Sincerely,

Cora Rose

Personal Reflection:

Throughout the development of this unit the hardest challenge I encountered was getting over the thought that just because I wasn't strictly teaching students a skill doesn't mean that they aren't learning and/or practicing said skill. The quote that has inspired me from the very beginning, through every assignment thus far (particularly in this mini unit) is that "Learning skills was not enough if it had no connection to the world and no direction of making it better" (Muhammad 2023). As I started to write out the lesson plans, I found myself falling into the trap of skill-based learning over and over again. However, having this quote in the back of my mind consistently reminded me that it is more beneficial for students when education is placed within a broader global context (which is what I set out this unit to do).

Another brief challenge I encountered in designing this unit was ensuring that all students felt represented, seen, and included in the material. I had to be intentional about choosing texts and examples that I utilized to reflect a range of identities and experiences. This is shown throughout my unit through the selection of whose engineering excerpts to show, the books I had students, read, and the websites I had students look at. Together they all aim to cover a range of students' identities and cultures.

I was further inspired by the statement of: "I tell the teachers whom I mentor that I see the world in curriculum. I can walk down the street and "see" lesson plans and how to implement them with youth. In this way, there is so much learning that can be historicized (when you look at the history of something) and intellectualized (when you connect a topic to other learning standards and intellect)" (Muhammad 2020). This quote shifted my mindset completely as I began to see my students' environment as full of possibilities for learning. This is where the idea of teaching a unit on renewable energy stemmed from. Renewable energy is not just a scientific topic for students to discuss, it is also a human and justice issue. Through this new lens this quote provided me I was able to more fully understand how to integrate science, reading, writing, and advocacy into a meaningful way in curriculum, rather than what I would have done prior through the separation of them into distinct "subjects." The use of the five pursuits really helped me accomplish the integration of science, reading, writing, and advocacy. For identity, I had students reflect on their role in their community and how energy use and environmental issues affect their lives. For skills, I have students practice reading nonfiction, writing persuasive letters, and engage in a multitude of discussions backed by evidence. For intellect, I have students thinking about how renewable energy works and how it connects to real-world systems. For criticality, I have students question who is most affected by environmental issues. Lastly for joy, I have students use their creativity to participate in hands on activities including building a windmill, mini posters, and advocacy letters to not only make learning fun but also more meaningful.

To pair with this, the idea that, "I define text as anything that can be read – both print texts and nonprint texts. Society members were reading print texts, but they were also reading the world as texts. They read images and the social times as texts" (Muhammad 2020) is also important. The way that I thought about literacy before was such a black and white view. Now I see literacy everywhere. In art, in pictures, in newspaper articles, on websites, and so many other places. By restricting students just to books as a teacher you are harming their ability to learn about how

literacy works in real life. As a whole, this mini unit has transformed the way I think about literacy. I will admit that prior to this class I tended to think about teaching literacy in a way that focused almost purely on whether students could fluently read or write structured paragraphs. However, now I see that literacy is so much more than this. Literacy is a social practice. Literacy is alive in the way that students communicate their ideas, connect with others, and (most importantly) how students make sense of the world. It is about actively creating and questioning information. It is not just about helping students become better readers and writers, it is more importantly about helping students become more empowered and thoughtful human beings. I feel like everything that I have learned can be briefly summed up with the following, "Teaching skills is important but teaching them alone is highly problematic and also should not be privileged over other goals or pursuits discussed in this book. Instead, skills should be taught alongside other pursuits" (Muhammad 2020). This assignment has shown me how to not only intertwine the five pursuits but also how each one enriches the others. You can't just teach skills in isolation and expect meaningful learning. Real literacy means helping students understand themselves, their world, and how to change it.

Going forward, I am committed to teaching practices that reflect the diversity of my students lived experiences. I will strive to choose texts (both print and non-print) that reflect multiple perspectives, create space for critical dialogue, and support my students in becoming readers, and writers, but also change-seekers and makers.

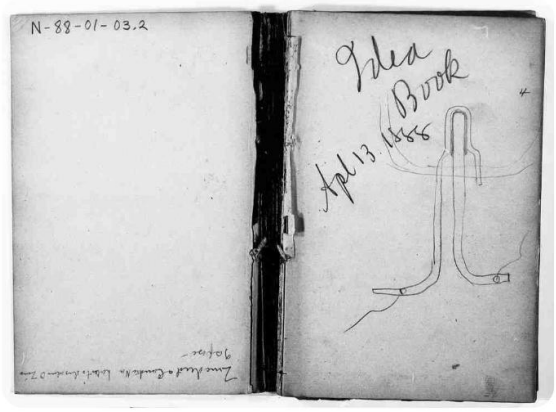
References:

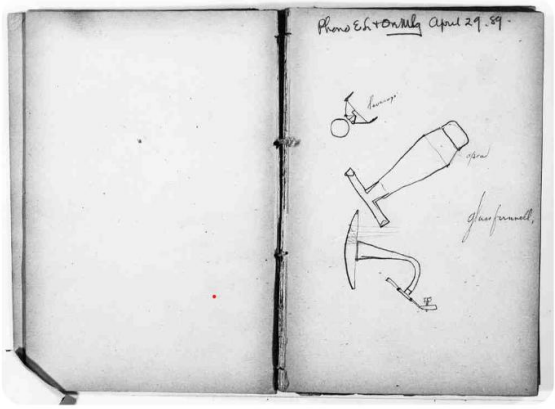
Muhammad, G. (2020). *Cultivating Genius: An Equity Framework for Culturally and Historically Responsive Literacy*. Scholastic Inc.

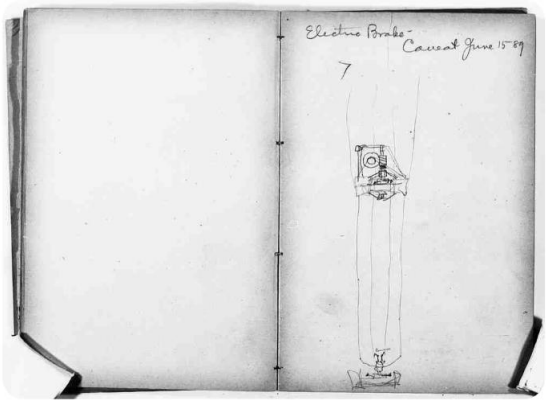
Muhammad, G. (2023). *Unearthing Joy: A Guide to Culturally and Historically Responsive Teaching and Learning*. Scholastic Professional.

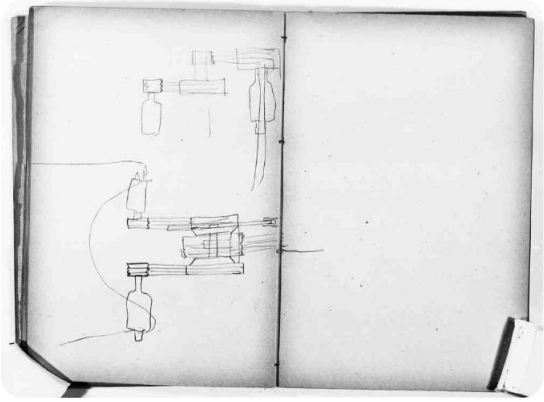
Lesson 1: Be an Inventor Like Edison!

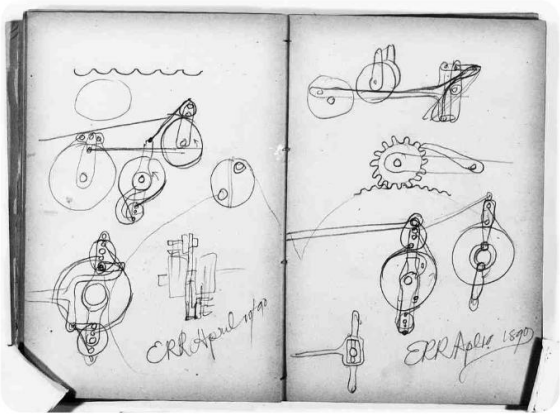
Appendix A: Thomas Edison Notebook Excerpts

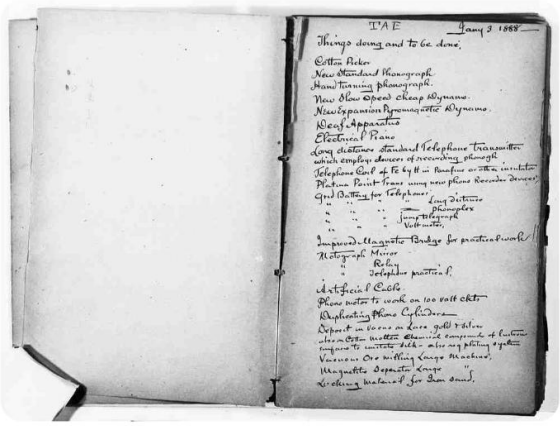


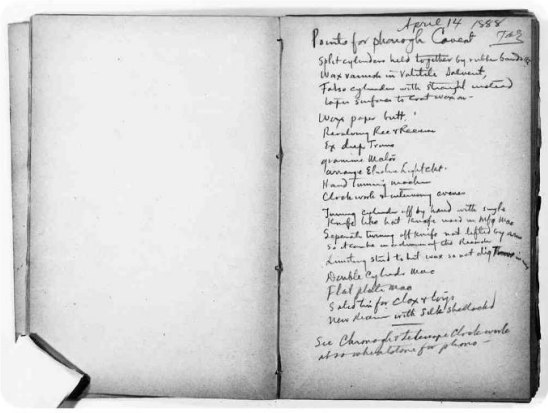


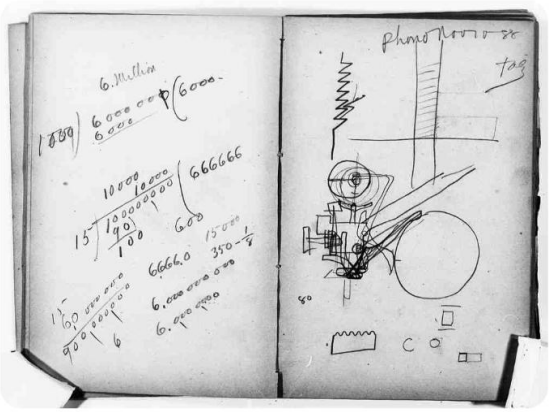


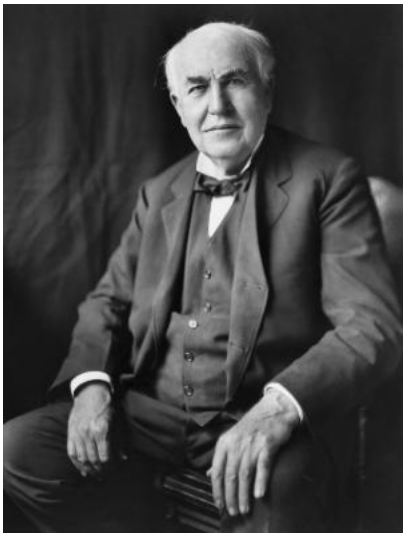






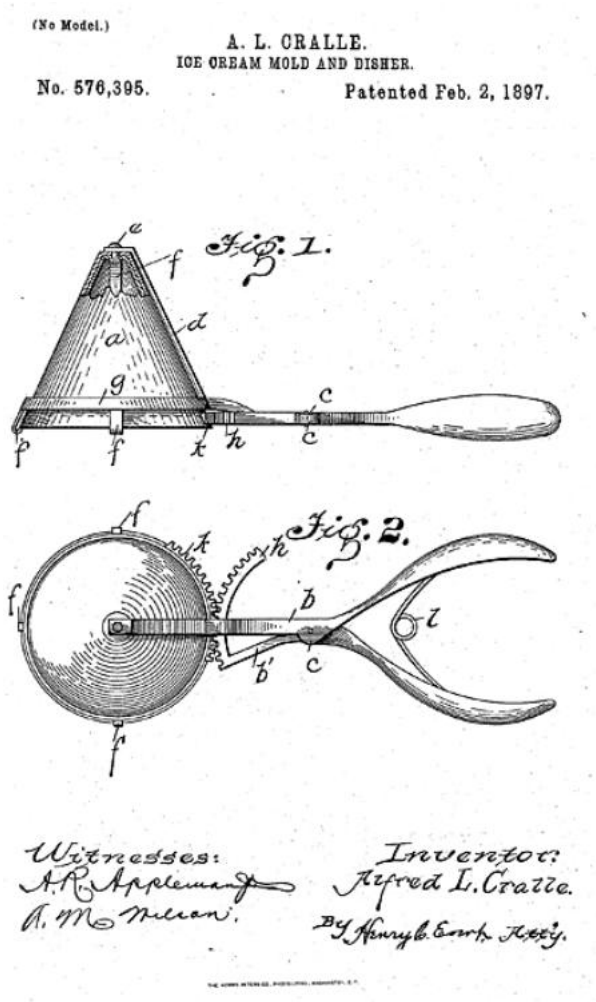

April 14 1888
Points for phonograph Caveat
Split cylinder held together by rubber bands
Wax varnish in volatile solvent,
False cylinder with thin foil instead
Lower surface to test wax on -
Wax paper butt.
Revolving Reservoir
Sp drop Tones
grooves Wales
Carriage Phelan & Collender
Hand Turning machine
Clock work stationary screw
Tone includes - off by hand with single
Knife like but knife used in mfg Wax
Separate turning off knife and lifter by arm
so stands on dimes of the Revolving
Limiting stud to lift knife is not slip Tones
Double cylinder wax
Flat plate wax
Select this for Clocks & toys
New drawn with Selbstschreiber
See Chronograph & Clock work
also which letters for phono -


6. Million
1889
6000
10000
666666
15
900
600
6666.0
350
6.000 000
6.000 000
phono
tag



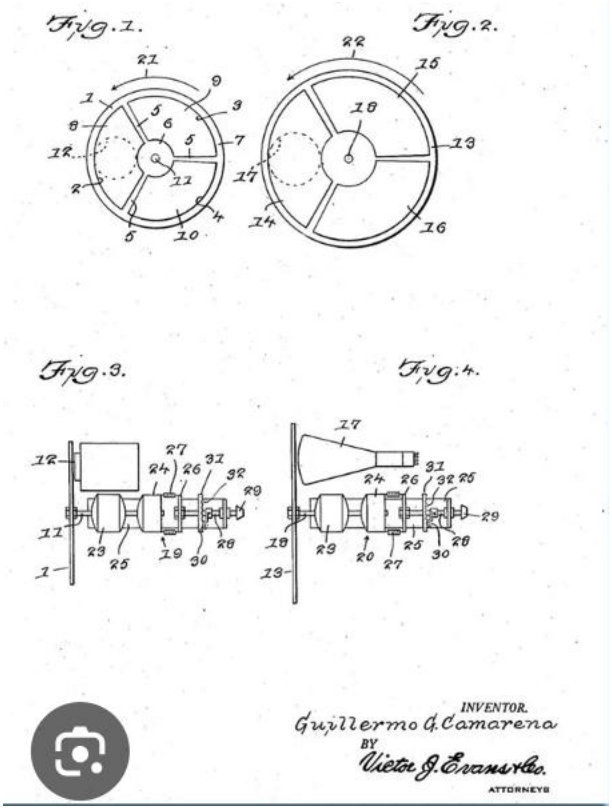
Lesson 2: Renewable Energy

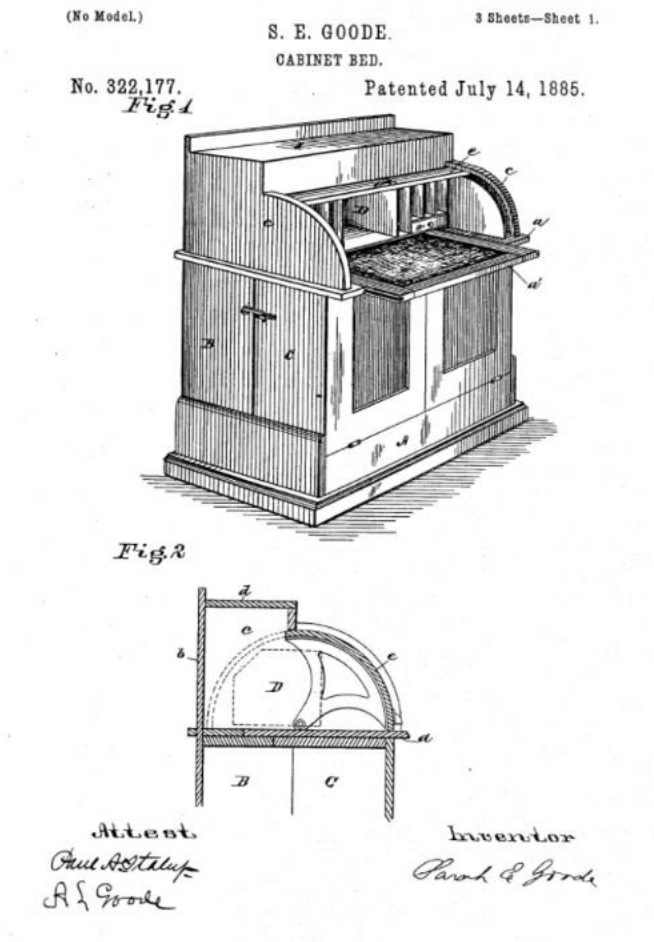
Appendix B: Alfred L. Cralle Ice Cream Scoop Idea



Lesson 3: Wind Energy Case Study

Appendix C: Guillermo González Camarena Color T.V Idea



Lesson 5: Engineering Challenge: Build a Windmill

Appendix D: Sarah Goode Cabinet-Bed idea